Supersingular Isogeny Diffie-Hellman Key Exchange

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1 Introduction

Most of the public key crypto-systems that we have discussed this term are dependent on the fact that there is no known efficient probabilistic classical algorithm of solving the integer factorization problem or the discrete logarithm problem.

Definition 1.1 (Integer Factorization Problem)**.** For any integer $N \in \mathbb{Z}$, find its decomposition into prime numbers $N = \prod_i p_i^{e_i}$.

By the fundamental theorem of arithmetic, integer factorization is unique up to reordering. The widely-used RSA crypto-system is constructed upon the assumption that classical factorization algorithms are infeasible.

Definition 1.2 (Discrete Logarithm Problem)**.** For any g and h in a finite group G, where $g^k = h$ and $k \in \mathbb{Z}$, find k.

Related to the discrete logarithm problem, there are computational and decisional Diffie-Hellman problems.

Definition 1.3 (Computational Diffie-Hellman Problem)**.** Given (g, g^x, g^y), where g is an element in a finite group G and $x, y \in \mathbb{Z}_+$, one cannot efficiently compute g^{xy}.

Definition 1.4 (Decisional Diffie-Hellman Problem)**.** Given (g^x, g^y, g^{xy}) and (g^x, g^y, g^z), where g is an element in a finite group G and $x, y, z \in \mathbb{Z}_+$, one cannot efficiently distinguish the two triplets.

The ElGamal cryto-system is based on the assumption that the Diffie-Hellman problem is intractable.

However, with the development of quantum computing theories, the difficulty of the integer factorization problem and the discrete logarithm problem are challenged. In a paper that he published in 1994, Peter Shor proved that the integer factorization and the discrete logarithm problem can be solved in the polynomial time with a quantum computer algorithm [1]. He showed that both the integer factorization problem and the discrete logarthm can be reduced to the order-finding problem.

Definition 1.5 (Order-Finding Problem)**.** For any $N \in \mathbb{Z}_+$ and $a \in \mathbb{Z}_N^\times$, find the smallest positive integer r such that $a^r \equiv 1 \pmod{N}$.

In his paper, Shor gave an efficient quantum algorithm that solves the order-finding problem in the polynomial time.

Although there is no quantum computer for practical use now, cryptographists have been devising new public key crytographic theories for which there is no known efficient quantum algorithm. In the NIST Post-Quantum Cryptography Competition, there are lattice-based, code-based, hash-based, multivariate, and isogeny algorithms. Supersingular isogeny Diffie-Hellman Key Exchange Algorithm (SIDH) is an quantum resistant cryptosystem from supersingular elliptic curve isogenies proposed in 2011 by Luca De Feo, David Jao, and Jérôme Plût [2].

2 Elliptic Curve Groups and Isogenies

First we want to define the elliptic curves. The following is the definition from algebraic geometry:

Definition 2.1 (Elliptic Curve)**.** An elliptic curve is a non-singular projective curve of genus 1 with a rational point.

Or we can write the definition in a more concrete form:

Definition 2.2 (Elliptic Curve)**.** An elliptic curve over the field F is a curve of the form $E : \{(x, y) \in F^2 \mid y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6\}$ where $a_1, \ldots, a_6 \in F$ and E is non-singular, that is, E has a well-defined tangent line at every point ($\frac{dx}{dy}|_P$ and $\frac{dy}{dx}|_P$ exists for any $P \in E$).

If we impose a restriction on the characteristics of the fields, we have a simpler form:

Definition 2.3 (Elliptic Curve)**.** If $\mathrm{Char}(F) \neq 2, 3$, then an elliptic curve over the field F has the form $E : \{(x, y) \in F^2 \mid y^2 = x^3 + ax + b\}$, where $a, b \in F$ and $4a^3 + 27b^2 \neq 0$.

In this paper, we only consider fields with characterstics not equal to 2 or 3.

Definition 2.4 (Elliptic Curve Group)**.** An elliptic curve group $E(F)$ over the field F is $E(F) = \{(x, y) \in F^2 \mid y^2 = x^3 + ax + b\} \cup \{\infty\}$, where $a, b \in F$, ∞ is the the identity element, and addition defined for $P = (x_P, y_P), Q = (x_Q, y_Q) \in E(F)$ is $P + Q = (m^2 - x_P - x_Q, -m(x_{P+Q} - x_P) - y_P)$, where $m = \frac{y_Q - y_P}{x_Q - x_P}$ if $P \neq Q$ and $m = \frac{3x_P^2 + a}{2y_P}$ if $P = Q$ with the exception that $(x, y) + (x, -y) = \infty$ where $x, y \in F$.

It is easy to verify that the addition is communtative, so the group is abelian.
Recall the Structure Theorem of Finite Abelian Groups:

Theorem 2.1 (Structure Theorem of Finite Abelian Groups)**.** *Any finite abelian group G is isomorphic to a direct product of cyclic groups of prime power orders: $G \cong \mathbb{Z}_{p_1^{k_1}} \times \cdots \times \mathbb{Z}_{p_n^{k_n}}$, where $n, k_1, \ldots, k_n \in \mathbb{Z}_+$ and $p_1, \ldots, p_n$ are primes,*

By this theorem, we can see that an elliptic curve group over a finite field is either cyclic or isomorphic to a direct product of cyclic groups of prime power orders. In fact, we have the following proposition:

Proposition 2.1. *An elliptic curve group over a finite field is either cyclic or isomorphic to a direct product of two cyclic groups.*

In order to prove this proposition, we define the n-torsion subgroup:

Definition 2.5 (n-torsion Subgroup)**.** For any $n \in \mathbb{Z}_+$ and elliptic curve $E(K)$ over the finite field K, we define $E[n] = \{P \in E(\overline{K}) \mid nP = \infty\}$, where $\overline{K}$ is the algebraic closure of K. It is easy to verify that $E[n]$ satisfies all group properties and is thus a group.

We have the following lemma related to the n-torsion subgroup:

Lemma 2.1. *Let E be an elliptic curve over $\mathbb{F}_p$, and assume that the prime p does not divide m. The there exists a value k such that $E(\mathbb{F}_{p^{jk}})[m] \cong \mathbb{Z}_m \times \mathbb{Z}_m$ for all $j \geq 1$. In particular if K is a finite field, we have $E(K)[m] \cong \mathbb{Z}_m \times \mathbb{Z}_m$*

Proof. For any elliptic curve group $E(F)$ over a finite field F, we have $E(F) \subset E[|E(F)|] = \mathbb{Z}_{|E(F)|} \times \mathbb{Z}_{|E(F)|}$, so $E(F)$ is either cyclic or a direct product of two cyclic groups. $\square$

The elliptic curve that we use to implement SIDH in this paper is supersingular, which is defined in the following way:

Definition 2.6 (Supersingular Elliptic Curve)**.** An elliptic curve is supersingular if its endomorphism ring is of rank 4 over $\mathbb{Z}$.

The other major definition that we need to construct SIDH is that of isogenies:

Definition 2.7 (Isogeny)**.** Let E and E' be elliptic curves over a field F. An isogeny between E and E' is a rational map $\phi : E \to E'$ such that ϕ is a non-constant group homomorphism.

In particular,

Definition 2.8 (Rational Map)**.** A rational map between curves $\phi : C \to C'$ is a map such that each coordinate of ϕ is a rational function (a quotient of polynomials).

Given an elliptic curve E and its subgroup G, Jacques Vélu showed in his paper in 1971 an algorithm to compute the unique isogeny with G as its kernel and the codomain curve of this isogeny. This algorithm is called Vélu's Formula [3].

Theorem 2.2 (Vélu's Formula)**.** *Consider a field K whose characteristic is neither 2 or 3 and an elliptic curve E over K of the form $y^2 = x^3 + ax + b$. Given a subgroup G of E, we define the map $\phi : E \to E'$ in the following way:*

- *If $P = (x_P, y_P) \notin G$, let*

$$\phi(P) = (x_P + \sum_{Q \in G \setminus \{\infty\}} (x_{P+Q} - x_Q), y_P + \sum_{Q \in G \setminus \{\infty\}} (y_{P+Q} - y_Q))$$

.

- *If $P \in G$, let $\phi(P) = \infty$.*

Compute $v = \sum_{P \in G \setminus \{\infty\}} (3x_P^2 + a)$ and $w = \sum_{P \in G \setminus \{\infty\}} (3x_P^3 + ax_P)$. The codomain curve E' is of the form: $y^2 = x^3 + Ax + B$, where $A = a - 5v$ and $B = b - 7w$. The map ϕ is the unique isogeny between E and E' with the kernel G.

3 The Key Exchange Protocol

The basic idea of SIDH is described as the follows:

1. Alice and Bob agrees on an elliptic curve E as the public key.
2. Alice chooses a subgroup $K_a \subset E$ and Bob chooses a subgroup $K_b \subset E$. K_a and K_b are the private keys.
3. Alice computes E_a and ϕ_a with Vélu's Formula and Bob computes E_b and ϕ_b with Vélu's Formula.
4. Alice sends $(E_a, \phi_a(K_b))$ to Bob and Bob sends $(E_b, \phi_b(K_a))$ to Alice.
5. Alice computes E_{ab} and ψ_a (the kernel is $\phi_b(K_a)$) with Vélu's Formula. Bob computes E_{ab} and ψ_b (the kernel is $\phi_a(K_b)$) with Vélu's formula. The shared secret between Alice and Bob is E_{ab}.

$$\begin{array}{ccc} E & \xrightarrow{\phi_a} & E_a \\ \downarrow{\scriptstyle \phi_b} & & \downarrow{\scriptstyle \psi_b} \\ E_b & \xrightarrow{\psi_a} & E_{ab} \end{array}$$

We need to show that Alice and Bob gets the same message E_{ab} in Step 5. We want to prove the following proposition:

Proposition 3.1. $\ker(\psi_b \circ \phi_a) = \ker(\psi_a \circ \phi_b) = \langle K_a \cup K_b \rangle$.

Proof. For any $P \in K_a \cup K_b$, we have $P \in K_a$ or $P \in K_b$. If $P \in K_a$, then $\psi_b \circ \phi_a(P) = \psi_b(\infty) = \infty$. If $P \in K_b$, then $\psi_b \circ \phi_a(P) = \psi_b(\phi_a(P)) = \infty$ since $\phi_a(P) \in \phi_a(K_b)$. Hence, $\langle K_a \cup K_b \rangle \subset \ker(\psi_b \circ \phi_a)$.

For any $P \in \ker(\psi_b \circ \phi_a)$, we have $\psi_b(\phi_a(P)) = \infty$, so $\phi_a(P) \in \ker \psi_b = \phi_a(K_b)$. We have $P \in K_b$ or $P \in \ker \phi_a = K_a$. Hence, $\ker(\psi_b \circ \phi_a) \subset \langle K_a \cup K_b \rangle$.

Therefore, we conclude $\ker(\psi_b \circ \phi_a) = \langle K_a \cup K_b \rangle$. Following the same argument, we can show that $\ker(\psi_a \circ \phi_b) = \langle K_a \cup K_b \rangle$ $\square$

Since $\psi_b \circ \phi_a$ and $\psi_a \circ \phi_b$ have the same kernel, by Vélu's Formula, they are the same isogeny and reach the same the codomain curve E_{ab}.

However, there are a few issues with this key exchange protocol. First, Alice does not know K_b, so she cannot compute $\phi_a(K_b)$. Similarly, Bob cannot compute $\phi_b(K_a)$. Moreover, anyone with the knowledge of $(E_a, \phi_a(K_b))$ and $(E_b, \phi_b(K_a))$ can compute E_{ab} with Vélu's Formula.

Theorem 3.1. *Consider the elliptic curve $E : y^2 = x^2 + ax + b$ defined over $\mathbb{F}_p$, where p is a prime that is greater than 3. Then $\phi(x, y) = (-x, iy)$ is an isomorphism from $E(\mathbb{F}_{p^2})$ to $E'(\mathbb{F}_{p^2})$ where E' is the curve $E' : y^2 = x^2 + ax - b$.*

Proof. First, we want to prove that $i = \sqrt{-1} \in \mathbb{F}_{p^2}$. It suffices to prove that there exists $g \in \mathbb{F}^*_{p^2}$ whose order is 4. Since $\mathbb{F}_{p^2}$ is a field, the multiplicative group $\mathbb{F}^*_{p^2}$ is cyclic of order $p^2 - 1$. Let h be the generator of this group and we know the order of h is $p^2 - 1$. Since $p > 3$, we have $p = 2k + 1$ for some $k \in \mathbb{Z}_+$. Let $g = h^{k^2+k}$. We can see that $g^4 = h^{4k^2+4k} = h^{p^2-1} = 1$. Since g is of order $p^2 - 1$, we know that $g^2 \neq 1$ and $g \neq 1$. Therefore, we have $i \in \mathbb{F}_{p^2}$.

Next, we want to show that ϕ is well-defined. For any $P = (x, y) \neq \infty$ in $E(\mathbb{F}_{p^2})$, we have $\phi(x, y) = (-x, iy)$, and $(-x)^3 + a(-x) - b = -(x^3 + ax + b) = -y^2 = (iy)^2$, so $\phi(x, y) \in E'(\mathbb{F}_{p^2})$.

Then we want to show that ϕ is a group homomorphism. If $P = (x, y), Q = (x, -y)$ for some $x, y \in \mathbb{F}_{p^2}$, we have $\phi(P + Q) = \phi(\infty) = \infty$, and $\phi(P) + \phi(Q) = (-x, iy) + (-x, -iy) = \infty = \phi(P + Q)$.

If $P = Q = (x, y)$ for some $x, y \in \mathbb{F}_{p^2}$, then $P + Q = (m^2 - 2x, -m(m^2 - 3x) - y) = (m^2 - 2x, -m^3 + 3mx - y)$, where $m = \frac{3x^2+1}{2y}$. We have $\phi(P + Q) = (-m^2 + 2x, -im^3 + 3imx - iy)$ and $\phi(P) + \phi(Q) = (-x, iy) + (-x, iy) = (n^2 + 2x, -n(n^2 + 3x) - iy) = (n^2 + 2x, -n^3 - 3nx - iy)$, where $n = \frac{3x^2+1}{2iy} = -im$, so $\phi(P) + \phi(Q) = (-m^2 + 2x, -im^3 + 3imx - iy) = \phi(P + Q)$.

If $Q = \infty$ (or equivalently $P = \infty$) and $P = (x, y)$ for some $x, y \in \mathbb{F}_{p^2}$, then $P + Q = (x, y)$. We have $\phi(P + Q) = \phi(x, y) = (-x, iy)$, $\phi(P) + \phi(Q) = (-x, iy) + \infty = (-x, iy) = \phi(P + Q)$.

Otherwise, if $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ for some $x_1, y_1, x_2, y_2 \in \mathbb{F}_{p^2}$, then $P + Q = (m^2 - x_1 - x_2, -m(m^2 - 2x_1 - x_2) - y_1) = (m^2 - x_1 - x_2, -m^3 + 2mx_1 + mx_2 - y_1)$, where $m = \frac{y_2 - y_1}{x_2 - x_1}$. We have $\phi(P + Q) = (-m^2 + x_1 + x_2, -im^3 + 2imx_1 + imx_2 - iy_1)$, and $\phi(P) + \phi(Q) = (-x_1, iy_1) + (-x_2, iy_2) = (n^2 + x_1 + x_2, -n(n^2 + 2x_1 + x_2) - iy_1) = (n^2 + x_1 + x_2, -n^3 - 2nx_1 - nx_2 - iy_1)$, where $n = \frac{iy_2 - iy_1}{-x_2 + x_1} = -im$, so $\phi(P) + \phi(Q) = (-m^2 + x_1 + x_2, -im^3 + 2imx_1 + imx_2 - iy_1) = \phi(P + Q)$.

Therefore, ϕ is a group homomorphism.

Moreover, we want to prove that ϕ is surjective. For any $Q = (x, y) \neq \infty$ in $E'(\mathbb{F}_{p^2})$, let $P = (-x, -iy)$. We can see that $(-x)^3 + a(-x) + b = -(x^3 + ax - b) = -y^2 = (-iy)^2$, so $P \in E(\mathbb{F}_{p^2})$ and $\phi(-x, -iy) = (x, y)$. Hence, ϕ is surjective.

Finally, we want to prove that ϕ is injective. We want to show that $\ker \phi$ consists only of the identity element. We want to find $P = (x, y) \in E(\mathbb{F}_{p^2})$ such that $\phi(P) = (-x, iy) = \infty$. The only P that satisfies this equation is ∞, and thus $\ker \phi = \{\infty\}$. Therefore ϕ is injective.

Therefore, ϕ is an isomorphism from $E(\mathbb{F}_{p^2})$ to $E'(\mathbb{F}_{p^2})$. $\square$

Remark. Let $\psi : E'(\mathbb{F}_{p^2}) \to E(\mathbb{F}_{p^2})$ be defined as $\psi(x, y) = (-x, -iy)$. We can easily verify that ψ is well-defined using the same argument as before. For any $P = (x, y) \neq \infty$ in $E(\mathbb{F}_{p^2})$, we have $\phi \circ \psi(x, y) = \phi(-x, -iy) = (x, y)$ and $\psi \circ \phi(x, y) = \psi(-x, iy) = (x, y)$, so $\psi \circ \phi$ and $\phi \circ \psi$ are identity. Therefore, we have an inverse of ϕ: $\phi^{-1} = \psi$.

Theorem 3.2. *$\pi(x, y) = (x^p, y^p)$ is a group homomorphism from $E(\mathbb{F}_{p^2})$ to $E(\mathbb{F}_{p^2})$.*

Proof. First we want to show that π is well-defined. The characteristic of the field $\mathbb{F}_{p^2}$ is p. For any $x, y \in \mathbb{F}_{p^2}$, we have $\pi(x,y) = (x^p, y^p)$ and $x^p, y^p \in \mathbb{F}_{p^2}$ (because $\mathbb{F}_{p^2}$ is a field). Since $\mathbb{F}_p^* \subset \mathbb{F}_{p^2}^*$ is cyclic, we have $g^{p-1} = 1$ for any $g \in \mathbb{F}_p^* \subset \mathbb{F}_{p^2}^*$. We can see that $(y^p)^2 = (y^2)^p = (x^3+ax+b)^p = x^{3p} + a^p x^p + b^p = (x^p)^3 + ax^p + b$, so $\pi(x,y) \in E(\mathbb{F}_{p^2})$.

If $P = (x,y), Q = (x,-y)$ for some $x, y \in \mathbb{F}_{p^2}$, we have $\pi(P+Q) = \pi(\infty) = \infty$, and $\pi(P)+\pi(Q) = (x^p, y^p) + (x^p, -y^p) = \infty = \pi(P+Q)$.

If $P = Q = (x,y)$ for some $x, y \in \mathbb{F}_{p^2}$, then $\pi(P+Q) = (m^{2p} - 2x^p, -m^{3p} + 3m^p x^p - y^p)$ and $\pi(P) + \pi(Q) = (x^p, y^p) + (x^p, y^p) = (t^2 - 2x^p, -t^3 + 3tx^p - y^p)$, where $t = \frac{3x^{2p}+1}{2y^p} = m^p$, so $\pi(P) + \pi(Q) = (m^{2p} - 2x^p, -m^{3p} + 3m^p x^p - y^p) = \pi(P+Q)$.

If $Q = \infty$ (or equivalently $P = \infty$) and $P = (x,y)$ for some $x, y \in \mathbb{F}_{p^2}$, then $P + Q = (x,y)$. We have $\pi(P+Q) = (x^p, y^p)$, and $\pi(P) + \pi(Q) = (x^p, y^p) + \infty = (x^p, y^p) = \pi(P+Q)$.

Otherwise, if $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ for some $x_1, y_1, x_2, y_2 \in \mathbb{F}_{p^2}$, then $P + Q = (m^2 - x_1 - x_2, -m(m^2 - 2x_1 - x_2) - y_1) = (m^2 - x_1 - x_2, -m^3 + 2mx_1 + mx_2 - y_1)$, where $m = \frac{y_2 - y_1}{x_2 - x_1}$. We have $\pi(P+Q) = (m^{2p} - x_1^p - x_2^p, -m^{3p} + 2m^p x_1^p + m^p x_2^p - y_1^p)$, $\pi(P) + \pi(Q) = (x_1^p, y_1^p) + (x_2^p, y_2^p) = (n^2 - x_1^p - x_2^p, -n(n^2 - 2x_1^p - x_2^p) - y_1^p) = (n^2 - x_1^p - x_2^p, -n^3 + 2nx_1^p + nx_2^p - y_1^p)$ where $n = \frac{y_2^p - y_1^p}{x_2^p - x_1^p} = m^p$, so $\pi(P) + \pi(Q) = (m^{2p} - x_1^p - x_2^p, -m^{3p} + 2m^p x_1^p + m^p x_2^p - y_1^p) = \pi(P+Q)$.

Therefore, π is an endomorphism of $E(\mathbb{F}_{p^2})$. ◻

Theorem 3.3. $|E(\mathbb{F}_p)| = p + 1 + \sum_{x \in \mathbb{F}_p} (\frac{x^3+ax+b}{p})$.

Proof. We have $p + 1 + \sum_{x \in \mathbb{F}_p} (\frac{x^3+ax+b}{p}) = 1 + \sum_{x \in \mathbb{F}_p} (1 + (\frac{x^3+ax+b}{p}))$.

Note that if $x_0^3 + ax_0 + b$ is a non-zero square modulo p, then there are 2 points with x-coordinate x_0 in $E(\mathbb{F}_p)$ and $1 + (\frac{x_0^3+ax_0+b}{p}) = 2$.

If $x_0^3 + ax_0 + b \equiv 0 \pmod p$, then there is one point with x-coordinate x_0 in $E(\mathbb{F}_p)$ and $1 + (\frac{x_0^3+ax_0+b}{p}) = 1$.

If $x_0^3 + ax_0 + b$ is a non-square modulo p, then there is no point with x-coordinate x_0 in $E(\mathbb{F}_p)$ and $1 + (\frac{x_0^3+ax_0+b}{p}) = 0$.

In addition, we have $\infty \in E(\mathbb{F}_p)$, so $|E(\mathbb{F}_p)| = p + 1 + \sum_{x \in \mathbb{F}_p} (\frac{x^3+ax+b}{p})$. ◻

Theorem 3.4. $|E(\mathbb{F}_{p^2})| = |E(\mathbb{F}_p)||E'(\mathbb{F}_p)|$.

Proof. First we want to prove that $\ker(1-\pi^2) = E(\mathbb{F}_{p^2})$. It is easy to see that ∞ is in both sets. We consider the non-identity elements. For any $(x,y) \in \ker(1-\pi^2)$, since $(x,y) = \pi^2(x,y) = ((x^p)^p, (y^p)^p) = (x^{p^2}, y^{p^2})$, we have $x^{p^2} = x$ and $y^{p^2} = y$, so $x, y \in \mathbb{F}_{p^2}$, and thus $\ker(1-\pi^2) \subset E(\mathbb{F}_{p^2})$. For any $(x,y) \in E(\mathbb{F}_{p^2})$, we have $\pi^2(x,y) = (x^{p^2}, y^{p^2}) = (x,y)$, so $(x,y) \in \ker(1-\pi^2)$, and thus $E(\mathbb{F}_{p^2}) \subset \ker(1-\pi^2)$. Therefore, we have $E(\mathbb{F}_{p^2}) = \ker(1-\pi^2)$.

Let $1 - \pi : E(\mathbb{F}_{p^2}) \to E(\mathbb{F}_{p^2})$ denote the map $(1-\pi)(P) = P - \pi(P)$. It is easy to see that ∞ is in both $E(\mathbb{F}_p)$ and $\ker(1-\pi)$. For any non-identity $P = (x,y) \in \ker(1-\pi)$, we have $P = \pi(P)$, so $x^p = x$ and $y^p = y$, and thus $x, y \in \mathbb{F}_p$. Hence, we have $\ker(1-\pi) \subset E(\mathbb{F}_p)$. For any non-identity $(x,y) \in E(\mathbb{F}_p)$, we have $\pi(x,y) = (x^p, y^p) = (x,y)$, so $(x,y) \in \ker(1-\pi)$, and thus $E(\mathbb{F}_p) \subset \ker(1-\pi)$. Therefore, we have $E(\mathbb{F}_p) = \ker(1-\pi)$.

Let $1 - \pi : E(\mathbb{F}_{p^2}) \to E(\mathbb{F}_{p^2})$ denote the map $(1+\pi)(P) = P + \pi(P)$. It is easy to see that ∞ is in both $\mathrm{im}(\phi^{-1}|_{E'(\mathbb{F}_p)})$ and $\ker(1+\pi)$. We consider the non-identity elements. For any $(-x, -iy) \in \mathrm{im}(\phi^{-1}|_{E'(\mathbb{F}_p)})$, where $(x,y) \in E'(\mathbb{F}_p)$, we have $\pi(-x,-iy) = (-x^p, (-i)^p y^p) = (-x, iy) = -(-x,-iy)$, so $(-x,-iy) \in \ker(1+\pi)$, and thus $\mathrm{im}(\phi^{-1}|_{E'(\mathbb{F}_p)}) \subset \ker(1+\pi)$. For any $(-x,y) \in \ker(1+\pi)$, we have $((-x)^p, y^p) = -(-x,y)$, so $(-x)^p = -x$ and $y^p = -y$. Let $y = -iz$. We see that $iz^p = (-i)^p z^p = (-iz)^p = iz$, so $z^p = z$, and thus $x, z \in \mathbb{F}_p$. Since $(-x)^3 + a(-x) + b = y^2$, we have $z^2 = (iy)^2 = -y^2 = x^3 + ax - b$, so $(x,z) \in E'(\mathbb{F}_p)$. We see $\phi^{-1}(x,z) = (-x,-iz) = (-x,y)$, so $(-x,y) \in \mathrm{im}(\phi^{-1}|_{E'(\mathbb{F}_p)})$, and thus $\ker(1+\pi) = \mathrm{im}(\phi^{-1}|_{E'(\mathbb{F}_p)})$. Therefore, we have $\mathrm{im}(\phi^{-1}|_{E'(\mathbb{F}_p)}) = \ker(1+\pi)$.

We want to show that $\ker[(1-\pi)\circ(1+\pi)] \cong E(\mathbb{F}_p) \times E'(\mathbb{F}_p)$. We know that $P \in \ker[(1-\pi)\circ(1+\pi)]$ if and only if $(1+\pi)(P) \in E(\mathbb{F}_p)$. Let $P = P_1 + P_2$, where $P_1 \in E(\mathbb{F}_p)$ and $P_2 \in \mathrm{im}(\phi^{-1}|_{E'(\mathbb{F}_p)})$. We have $(1+\pi)(P_1+P_2) = (1+\pi)(P_1) + (1+\pi)(P_2) = (1+\pi)(P_1) \in E(\mathbb{F}_p)$. In addition, for any given

P, P_1, let $Q = \phi(P - P_1)$. We see that $P_2 = P - P_1 = \phi^{-1}(Q)$, so $P_2 \in \operatorname{im}(\phi^{-1}|_{E'(\mathbb{F}_p)})$. There are $|E(\mathbb{F}_p)||\operatorname{im}(\phi^{-1}|_{E'(\mathbb{F}_p)})| = |E(\mathbb{F}_p)||E'(\mathbb{F}_p)|$ such P's, and since $1 - \pi^2 = (1 - \pi) \circ (1 + \pi)$, we have $|E(\mathbb{F}_{p^2})| = |\ker(1 - \pi^2)| = |\ker[(1 - \pi) \circ (1 + \pi)]| = |E(\mathbb{F}_p)||E'(\mathbb{F}_p)|$. $\square$

Now consider the elliptic curve $E : y^2 = x^3 + x$ over $\mathbb{F}_p$, where p is a prime of the form $p = 2^e \cdot 3^f - 1$ and $e > 1$. Note that $p \equiv 3 \pmod 4$. By Theorem 3.1, we know that ϕ is an automorphism of $E(\mathbb{F}_{p^2})$.

Proposition 3.2. *If $n \in \mathbb{N}$ is odd and $P \in E[n] \cap E(\mathbb{F}_p)$, then $\phi(P)$ generates a subgroup of $E[n]$ disjoint from the subgroup generated by P, namely, their intersection consists only of the identity element.*

Proof. Since $P \in E[n] \cap E(\mathbb{F}_p)$, the subgroup generated by P is a subgroup of $E(\mathbb{F}_p)$. Since $p \equiv 3 \pmod 4$, we have $(\frac{-1}{p}) = -1$, so -1 is quadratic non-residue in $\mathbb{F}_p$ and thus $i = \sqrt{-1} \notin \mathbb{F}_p$. For any non-identity element $a\phi(P)$ in the subgroup generated by $\phi(P)$, where $a \in \mathbb{Z}$, we have $a \cdot \phi(P) = \phi(a \cdot P) = (-x_0, iy_0)$ where $a \cdot P = (x_0, y_0) \in E(\mathbb{F}_p)$. If $a \cdot P = (0, 0)$, then $2a \cdot P = \infty$ and thus $n \mid (2a)$. Since n is odd, we have $n \mid a$, then $a \cdot P = \infty$, which is a contradiction. If $a \cdot P \neq (0, 0)$, since $i \notin \mathbb{F}_p$, we know $a\phi(P) \notin E(\mathbb{F}_p)$. Therefore, intersection of the subgroup generated by P and the subgroup generated by $\phi(P)$ is the identity element, and thus they are disjoint. $\square$

Remark. If n is even, the subgroup generated by P and the subgroup generated by $\phi(P)$ are not necessarily disjoint. For example, if $n = 2$, we can see that $(0, 0) + (0, 0) = \infty$, so $(0, 0) \in E[2] \cap E(\mathbb{F}_p)$. Let $P = (0, 0)$, and we have $\phi(P) = (0, 0)$. Since the non-identity element $(0, 0)$ is in both the subgroup generated by P and the subgroup generated by $\phi(P)$, they are not disjoint. However, if $P \neq (0, 0)$, then the argument in the previous proof still works, and the subgroup generated by P and the subgroup generated by $\phi(P)$ are disjoint.

Proposition 3.3. $|E(\mathbb{F}_p)| = 2^e \cdot 3^f$.

Proof. We observe that $(\frac{(-x)^3 + (-x)}{p}) = (\frac{-1}{p})(\frac{x^3 + x}{p}) = -(\frac{x^3 + x}{p})$. We can see that $-\sum_{x \in \mathbb{F}_p}(\frac{x^3 + x}{p}) = \sum_{x \in \mathbb{F}_p}(\frac{(-x)^3 + (-x)}{p}) = \sum_{x \in \mathbb{F}_p}(\frac{x^3 + x}{p})$, so $\sum_{x \in \mathbb{F}_p}(\frac{x^3 + x}{p}) = 0$. By Theorem 3.3, we have $|E(\mathbb{F}_p)| = p + 1 = 2^e \cdot 3^f$. $\square$

Remark. By Theorem 3.4, we can see that $|E(\mathbb{F}_{p^2})| = |E(\mathbb{F}_p)|^2 = 2^{2e} \cdot 3^{2f}$. Since $E(\mathbb{F}_p) \subset E(\mathbb{F}_{p^2})$ and ϕ is an isomorphism, we have $E(\mathbb{F}_{p^2}) \cong E(\mathbb{F}_p) \times E(\mathbb{F}_p)$. If $E(\mathbb{F}_p)$ is not cyclic, then it is a direct product of two cyclic groups, and then $E(\mathbb{F}_{p^2})$ cannot be expressed as a direct product of two cyclic groups, which is a contradiction. Therefore, $E(\mathbb{F}_p)$ has to be cyclic.

By Lemma 2.1, we know that $E[2^e] \cong \mathbb{Z}_{2^e} \times \mathbb{Z}_{2^e}$ and $E[3^f] \cong \mathbb{Z}_{3^f} \times \mathbb{Z}_{3^f}$. Since $E(\mathbb{F}_p)$ is cyclic, ff we choose a point P at random in $E(\mathbb{F}_p)$, there is a high probability that P has order $2^e \cdot 3^f$. Repeat this process until we get a P with order $2^e \cdot 3^f$. Let $P_a = 3^f P$ and $P_b = 2^e P$, and we can see that the order of P_a is 2^e and the order of P_b is 3^f. Let $Q_a = \phi(P_a)$ and $Q_b = \phi(P_b)$. We have the following proposition:

Proposition 3.4. *$\{P_a, Q_a\}$ generates $E[2^e]$ and $\{P_b, Q_b\}$ generates $E[3^f]$.*

Proof. We can see that $2^e Q_a = \phi(2^e P_a) = \infty$, so $Q_a \in E[2^e]$. The order of Q_a is 2^e, because if $2^t Q_a = \infty$ for some $t < e$, then $\phi(2^t P_a) = \infty$, so $2^t P_a = \infty$ (because ϕ is an isomorphism), which is a contradiction. By Proposition 3.2, the subgroup generated by P_a and the subgroup generated by Q_b are disjoint, so $|\langle P_a \rangle \oplus \langle Q_a \rangle| = 2^e \cdot 2^e = |E[2^e]|$. Therefore, we have $E[2^e] = \langle P_a \rangle \oplus \langle Q_a \rangle$. Similarly, we have $E[3^f] = \langle P_b \rangle \oplus \langle Q_b \rangle$. $\square$

Instead of choosing specific kernel K_a and K_b as the private key in Step 2, Alice and Bob can agree on E, P_a, P_b as the public key in Step 1, and the Alice chooses $\alpha_a \in \mathbb{Z}$ as her private key and Bob chooses $\alpha_b \in \mathbb{Z}$ as his private key. Let $K_a = \langle P_a + \alpha_a Q_a \rangle$ and $K_b = \langle P_b + \alpha_b Q_b \rangle$. We have the revised protocol:

1. Alice and Bob agrees on a public key consisting of an elliptic curve $E : y^2 = x^3 + x$ over $\mathbb{F}_p$, $P_a \in E(\mathbb{F}_p)$ of order 2^e, and $P_b \in E(\mathbb{F}_p)$ of order 3^f. Compute $Q_a = \phi(P_a)$ and $Q_b = \phi(P_b)$.

2. Alice chooses $\alpha_a \in \mathbb{Z}$ and Bob chooses $\alpha_b \subset \mathbb{Z}$. α_a and α_b are the private keys. Alice computes $K_a = \langle P_a + \alpha_a Q_a \rangle$ and Bob computes $K_b = \langle P_b + \alpha_b Q_b \rangle$.

3. Alice computes E_a and ϕ_a with Vélu's Formula and Bob computes E_b and ϕ_b with Vélu's Formula.

4. Alice sends $(E_a, \phi_a(P_b), \phi_a(Q_b))$ to Bob and Bob sends $(E_b, \phi_b(P_a), \phi_b(Q_a))$ to Alice.

5. Alice computes E_{ab} and ψ_a (the kernel is $\phi_b(K_a) = \langle \phi_b(P_a) + \alpha_a \phi_b(Q_a) \rangle$) with Vélu's Formula. Bob computes E_{ab} and ψ_b (the kernel is $\phi_a(K_b) = \langle \phi_a(P_b) + \alpha_b \phi_a(Q_b) \rangle$) with Vélu's formula. The shared secret between Alice and Bob is E_{ab}.

The revised protocol does not have the issues in the original protocol. In the Step 4, Alice is able to communicate $\phi_a(K_b)$ to Bob without knowing K_b. Similarly, Bob is able to communicate $\phi_b(K_a)$ to Alice without knowing K_a. Someone with the knowledge of $(E_a, \phi_a(P_b), \phi_a(Q_b))$ and $(E_b, \phi_b(P_a), \phi_b(Q_a))$ is not able to construct E_{ab}.

The security of SIDH does not rely on the difficulty of integer factorization problem or the discrete logarithm problem. In fact, the discrete logarithm problem on $E[2^e]$ and $E[3^f]$ are easy if we consider the Weil Pairing defined on these torsion groups. The Weil Pairing has a long-winded definition, which can be referred to in [4], and has the following properties:

Properties 3.1 (Weil Pairing Properties)**.** *1. The values of the Weil Pairing satisfy $e_m(P,Q)^m = 1$.*

2. The Weil Pairing is bilinear, which means that $e_m(P_1+P_2, Q) = e_m(P_1, Q) + e_m(P_2, Q)$ and $e_m(P, Q_1 + Q_2) = e_m(P, Q_1) + e_m(P, Q_2)$ for any $P, P_1, P_2, Q, Q_1, Q_2 \in E[m]$.

3. The Weil Pairing is alternating, which means that $e_m(P,P) = 1$.

4. The Weil Pairing is non-degenerate, which means that if $e_m(P,Q) = 1$ for all $Q \in E[m]$, then $P = \infty$.

Proposition 3.5. *the discrete logarithm problem on $E[2^e]$ and $E[3^f]$ can be solved efficiently.*

Proof. For any element $R \in E[2^e]$, we can find $\alpha, \beta \in \mathbb{Z}$ such that $R = \alpha P_a + \beta Q_a$. We have $1 = e_{2^e}(\alpha P_a + \beta Q_a, R) = \alpha e_{2^e}(P_a, R) + \beta e_A(Q_a, R)$. Since $e_{2^e}(P_a, R)$ and $e_{2^e}(Q_a, R)$ can be efficiently computed, we just need to find the solution of a linear equation, which can be efficiently solved. Therefore, the generalized discrete logarithm problem is easy in $E[2^e]$. Following the same argument, we can see that the generalized discrete logarithm problem is easy in $E[3^f]$. $\square$

The security of this key exchange protocol relies on the difficulty of computing the isogeny. For example, one algorithm of computing the isogeny, Vélu's Formula, is highly inefficient. For any K_a generated by $P_a + \alpha_a Q_a$, we can see that $2^e(P_a + \alpha_a Q_a) = 2^e P_a + \alpha_a(2^e Q_a) = \infty$, so $|K_a| \leq 2^e$. There are steps in Vélu's Formula that operate on every non-identity element in the kernel. In this way, a brute-force search for α_a and α_b is not practical. There is no known quantum polynomial-time algorithm of finding isogenies yet, so this key exchange protocol is currently quantum resistant.

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